

Department of Computer Science and
Engineering,
School of Technology,
Pandit Deendayal Energy University
(PDEU)
Software Engineering(24CS209P)

Lab Experiment - 2

Name: Dax B Patel

Roll no: 24BCP269

Functional Requirements

- The system shall allow the librarian to add, update, and delete book records.
- The system shall maintain detailed information for each book including title, author, ISBN, publisher, and category.
- The system shall allow librarians to register, update, and delete library member records.
- The system shall support issuing books to registered members.
- The system shall record book return transactions.
- The system shall automatically update and track book availability status.
- The system shall calculate late return fines based on predefined library rules.
- The system shall allow users to search books by title, author, ISBN, or category.
- The system shall generate reports for issued books, returned books, and overdue books.
- The system shall maintain a complete history of all book transactions.

Non-Functional Requirements

- The system shall provide a user-friendly and intuitive interface for all users.
- The system shall ensure data security using role-based access control.
- The system shall respond to user actions within acceptable time limits.
- The system shall be reliable and available during library working hours.
- The system shall support data backup and recovery mechanisms.

User Stories

- As a librarian, I want to add and update book details so that the book catalog remains accurate.
- As a librarian, I want to manage member records so that only authorized users can borrow books.
- As a librarian, I want to issue books to members so that borrowing activities are properly tracked.
- As a librarian, I want to record book returns so that book availability is updated correctly.
- As a librarian, I want the system to calculate fines automatically so that penalties are consistent and fair.
- As a library member, I want to search for books so that I can quickly find required resources.
- As a librarian, I want to view overdue books so that I can follow up with members.
- As a librarian, I want to generate reports so that library usage and performance can be analyzed.
- As a librarian, I want to track book availability so that inventory is managed efficiently.
- As an administrator, I want to control access levels so that system data remains secure.

USECASE DIAGRAM

